

Government-Guaranteed Lending in Crisis: Lessons from Chile’s Credit Guarantee Program

Cristián Sánchez*

Universidad de los Andes

September 7, 2026

Abstract

Government-guaranteed lending was a central policy response to the COVID-19 shock. This paper synthesizes the evidence on Chile’s pandemic expansion of its public credit-guarantee scheme—the Small and Medium Enterprise Guarantee Fund, or FOGAPE-COVID—and organizes it around five design levers: targeting and eligibility, risk-sharing and pricing, bank incentives and oversight, coordination with complementary policies, and repayment and exit. Three results emerge from reading the studies jointly. First, the parameters that govern fiscal exposure are the exclusion screen and the loan-size cap rather than the headline coverage rate: relaxing the default-based eligibility screen would have raised expected losses by a factor of 2.7, and a 2.5% first-loss deductible was the difference between the government bearing 41% and 77% of those losses. Second, whether the program prevented defaults or merely deferred them is unresolved: the two studies of the crisis program that address the question reach opposite conclusions. I show that these differences follow from comparison groups, horizons, and crisis-specific screens and caps rather than from the data, with direct implications for how such programs should be wound down. Third, the repayment phase, not the disbursement phase, is where the Chilean evidence is thinnest and where the international record is most cautionary. The paper concludes with a research agenda for crisis lending.

Keywords: Credit guarantees; Crisis lending; Small businesses; Banks; Chile; Defaults

JEL classification: G21; G28; H81; E58; O16

*Corresponding author. School of Business and Economics, Universidad de los Andes. Monseñor Álvaro del Portillo 12455, Las Condes, Región Metropolitana, Chile. E-mail: cristiansm@gmail.com. All errors are my own.

1 Introduction

Government-guaranteed lending became a central policy tool during the COVID-19 pandemic, especially for small and medium enterprises (SMEs) facing abrupt revenue losses and tightening liquidity. In Chile, this response took the form of a large expansion of the existing Small and Medium Enterprise Guarantee Fund (FOGAPE), commonly referred to as FOGAPE-COVID. The program was deployed at speed and at scale: guaranteed loans accounted for 45% of all new loans by May 2020 (Chittaro and Sánchez, 2026), and roughly a quarter of eligible firms had taken a guaranteed loan by the end of that year (Huneus et al., 2025).

Chile is now an unusually well-documented case. Because the Financial Market Commission (CMF) and the Central Bank of Chile were able to link loan applications, bank approvals and rejections, the credit registry, and administrative tax records for the universe of formal firms, the program has been studied with research designs—eligibility discontinuities, lender-leniency instruments, and randomized offers—that are rarely available for a crisis intervention anywhere. That density of evidence is what makes a synthesis worth writing. It is also what makes one necessary, because the studies do not agree.

This paper reviews the accumulated evidence on FOGAPE and FOGAPE-COVID and organizes it around the design choices a policymaker actually controls. It makes three contributions that no individual study provides.

First, it *reconciles conflicting estimates*. The most policy-relevant question about crisis guarantees—whether they prevent defaults or merely defer them—is answered in opposite ways by the two studies of FOGAPE-COVID that address it. Rishmawi and Rojas (2026) find that access to FOGAPE-COVID reduced firms’ delinquency probability by about 1 percentage point in the short run and 4 points in the medium run, with the effect persisting for more than two years. Chittaro and Sánchez (2026) find that default risk fell sharply through 2021 and then converged, concluding that the program postponed rather than prevented defaults. The evidence on the standing program in normal times points in yet a third direction: Cowan et al. (2015) find that the pre-pandemic FOGAPE *worsened* borrowers’ payment behavior. Section 5 shows that these results are not contradictory once their estimands and settings are distinguished, and that what separates them—comparison group, horizon, and whether the crisis-specific eligibility screen and rate cap were in force—is itself informative about design.

Second, it *documents the state of the regional evidence base*. Section 4 reports the results of a systematic search for evaluations of Latin American pandemic guarantee schemes. Only one peer-reviewed evaluation of a non-Chilean Latin American COVID guarantee program exists (Pardo-Figueroa, 2026); no peer-reviewed firm-level econometric evaluation of Brazil’s

PRONAMPE or PEAC-FGI has appeared; and the best regional comparative source (Rudolph et al., 2022) is a descriptive survey of scheme administrators without microdata. This scarcity is a finding in its own right. It bounds how far any set of design lessons can travel, and it explains why this review is organized around the Chilean evidence rather than presented as a regional synthesis.

Third, it *maps design levers to identified mechanisms*. Each study identifies one link between a program parameter and an outcome; none assembles the full mapping. Section 5 organizes the evidence around five levers—targeting and eligibility; risk-sharing and pricing; bank incentives and oversight; coordination with complementary policies; and repayment and exit—and, for each, separates what the studies agree on from what they contest and what remains unmeasured.

Three themes emerge. Design parameters that look technical are first-order for fiscal exposure: removing the default-based eligibility screen would have raised expected losses by a factor of 2.7, and removing deductibles would have shifted the government’s share of expected loss from 41% to 77% (Huneus et al., 2025). Banks are active agents rather than conduits, and the direction of their selection appears to depend on whether screens and rate caps bind. And the repayment phase, not the disbursement phase, is where the Chilean evidence is thinnest and where the international record is most cautionary.

The remainder proceeds as follows. Section 2 describes FOGAPE and its pandemic expansion. Section 3 states the review’s scope, inclusion criteria, and corpus, and flags the publication status of every study. Section 4 places Chile’s design in regional and international perspective. Section 5 presents the evidence lever by lever. Section 6 states the limitations and the conditions under which these lessons generalize. Section 7 concludes with a research agenda.

2 Chile’s Pandemic Expansion of Its Public Guarantee Fund

Chile’s Small and Medium Enterprise Guarantee Fund (FOGAPE) was created in 1980. FOGAPE guarantees loans, leasing operations, and other financial products that public and private financial institutions offer to small and medium-sized firms. The primary funding sources of FOGAPE are government contributions, fees for guarantee services, and returns on fund investments. Banco Estado, Chile’s main public bank, manages the program, and the Financial Market Commission (CMF), the country’s financial regulator, supervises it.

FOGAPE allocates guarantees to banks, who decide on firm applications based on their internal criteria. Over time, the program has adjusted the borrower’s eligibility, coverage rates, and usage restrictions.

In response to the COVID-19 pandemic, the Chilean government expanded FOGAPE’s size and scope to enhance firms’ access to financial markets. On April 24, 2020, Law 21,229 temporarily modified FOGAPE, creating the pandemic-era expansion referred to as FOGAPE COVID. This reform increased the fund’s capital by USD 3 billion, sharply raising the volume of guarantees the fund could issue.¹ The expansion also extended eligibility to medium-sized and some large firms (up to UF 1,000,000 in sales), which had traditionally been ineligible.² To qualify for the program, firms must meet the following eligibility requirements: they cannot have overdue loan payments exceeding 30 days within the banking system, they must not have an individual risk classification of C1 or lower, and they cannot be undergoing reorganization or liquidation.³ As Section 5 shows, this last set of restrictions—the “do-not-finance” margin—turns out to be among the most consequential parameters in the entire design.

The pandemic expansion established several conditions for loans granted with government guarantees. These include a maximum interest rate of the monetary policy rate (MPR) plus 3%, a limit on the total amount of guaranteed loans of one quarter of the firm’s annual sales, financing restricted to working capital only, a grace period of at least six months, and loan terms between 24 and 48 months.⁴ The program stipulates that guarantee rates are determined by firm size, with smaller firms receiving higher guarantee rates. There are five size categories: micro (below UF 2,400 in annual sales), small (UF 2,400–25,000), medium (UF 25,000–100,000), large I (UF 100,000–600,000), and large II (UF 600,000–1,000,000), which map into four guarantee rates because micro and small firms are treated identically. The guarantee rate for loans to micro and small firms is 85%. The guaranteed rate for loans to medium-sized firms is 80%, for large I firms it is 70%, and for large II firms it is 60%.

¹Accounts of the resulting guarantee envelope differ across the studies reviewed here: [de Elejalde and Sánchez \(2025\)](#) report a capacity of up to USD 33 billion, or approximately 10% of GDP, while [Acosta-Henao et al. \(2024\)](#) and [Huneus et al. \(2025\)](#) report up to USD 24 billion. Throughout this paper I distinguish the guarantee *envelope*—the maximum the fund was authorized to guarantee—from credit *actually disbursed*, which was 4.6% of GDP ([Huneus et al., 2025](#)). Statements that the program was “worth ten percent of GDP” refer to the former and should not be read as a measure of realized support.

²UF is a unit of account used to adjust values for inflation in Chile. As of October 2025, 1 UF is approximately USD 40.

³Banks must assign a risk rating to borrowers when their exposure to the borrower is significant. Borrowers with overdue payments exceeding 90 days or those likely to require debt restructuring receive an individual risk rating ranging from C1 to C6.

⁴In April 2020, the MPR was 0.5%, resulting in an effective interest rate of 3.5%, which was favorable compared to market rates for most firms.

The program also includes a deductible for banks on each size-specific portfolio of guaranteed loans, designed to ensure that banks internalize some of the costs in case of default. Initially, these deductibles were higher for smaller firms, likely due to their higher default risk. However, as of June 30, 2020, all firm sizes were subject to a standardized deductible of 2.5%. The deductible significantly impacts the effective guarantee rates. For example, for small firms with an 85% guarantee and a 2.5% deductible, a 20% default rate results in an effective coverage rate of 75%. Table 1 below presents the guarantee rates and deductibles by firm size.

Table 1: Guarantee Rates and Deductibles by Firm Size

Type of firms	Net annual sales (UF)		Guarantee rate (%)	Deductible (%)
	Above	Below		
Micro and small	0	25,000	85	5 (2.5 after 6/30/2020)
Medium	25,000	100,000	80	3.5 (2.5 after 6/30/2020)
Large I	100,000	600,000	70	2.5
Large II	600,000	1,000,000	60	2.5

Notes: This table reports guarantee rates and deductibles for Chile’s pandemic guarantee program by firm size.

The program’s reach was large and front-loaded. Guaranteed loans accounted for 45% of all new loans by May 2020 (Chittaro and Sánchez, 2026), and guaranteed flows were concentrated in May and June 2020, at about 3% and 2% of GDP respectively (Albagli et al., 2023). By the end of 2020, nearly 250,000 firms—about 40% of firms reporting positive sales in February 2020—had obtained at least one loan through the program (Albagli et al., 2023). Financing conditions were far more favorable than the market alternative: the capped 3.5% rate compares with roughly 9% on contemporaneous non-guaranteed loans, and about three quarters of guaranteed loans carried the maximum 48-month maturity (Huneus et al., 2025).

Who the program served depends on whether one counts operations or pesos. Measured by the number of guarantees allocated, micro and small firms took the lion’s share (Rudolph et al., 2022); measured by volume, most pesos flowed to larger, safer firms, which account for a larger share of aggregate sales (Huneus et al., 2025). This gap between the count and volume distributions recurs throughout the review: statements about the program’s allocation depend heavily on which of the two one measures.

One further aggregate fact frames what follows. Albagli et al. (2023), using merged administrative records for the universe of formal firms, document that the leverage of the

median firm with positive bank debt roughly tripled, rising by more than 9 percentage points of 2019 sales. That accumulation of debt is the mechanism through which a liquidity intervention in 2020 becomes a solvency question in 2022, and it frames Section 5’s discussion of the repayment phase.

This institutional overview identifies the policy “knobs” embedded in the program—eligibility rules, caps on rates and loan sizes, partial guarantees, and deductibles—that the rest of the paper connects to observed outcomes.

3 Scope, Method, and the Evidence Base

3.1 Scope and inclusion criteria

This is a synthesis of the evidence on one program. The primary corpus consists of studies that estimate effects of FOGAPE or FOGAPE-COVID on firm or loan outcomes using Chilean microdata. Studies of the pre-pandemic scheme are included as normal-times benchmarks: they estimate what the standing guarantee fund does to credit access and repayment incentives absent the crisis expansion’s screens, caps, and scale, and are read as such rather than as evidence on FOGAPE-COVID itself. Studies of other countries’ schemes are used in two bounded ways: to benchmark design parameters (Section 4) and, within each design lever, to supply published evidence on mechanisms for which the Chilean record is thin or rests on a single unpublished source. They are not treated as evidence about Chile.

Studies were identified by searching Crossref, RePEc/IDEAS, SSRN, and OpenAlex for FOGAPE and for Chilean credit-guarantee terms in English and Spanish, and by hand-searching the working paper series of the Central Bank of Chile, the CMF, DIPRES, the World Bank, the IDB, and the NBER. Searches were run through August 2026. Comparative regional coverage was assembled by repeating this procedure for Peru, Brazil, Colombia, Mexico, and Argentina. Studies were included if they report quantitative estimates for identified firm or loan populations; institutional and legal descriptions were retained only as sources of program parameters.

Two features of the resulting corpus require explicit treatment. First, it is small: the program has been evaluated by roughly a dozen studies, several of which share data and authors and are therefore not independent. Second, it is predominantly grey literature. Of the FOGAPE evaluations located, only Cowan et al. (2015) and Rishmawi and Rojas (2026) have completed peer review; one further study, Huneus et al. (2025), is under revision at the *American Economic Journal: Macroeconomics* but is treated as unpublished here. Table 2

therefore records the publication status of each study, and the discussion returns to it where it affects how much weight a finding can bear—most notably for the bank-reclassification result in Section 5.

3.2 The corpus

Table 2 lists the corpus: for each study, the period covered, the research design, the main outcomes, the horizon over which effects are measured, and publication status as of August 2026. Three features of this corpus shape how it should be read. First, the designs answer different questions: a within-firm comparison of insured and uninsured loans, a demand-conditioned comparison of approved and rejected applicants, and a lender-leniency instrument recover different estimands, which Section 5.5 shows is central to reconciling their conclusions. Second, horizons range from a few months to five years, and several of the disagreements discussed below sharpen or dissolve once horizons are aligned. Third, the studies are not statistically independent: most draw on the same CMF and Central Bank administrative records, and several share authors, so agreement among them is weaker corroboration than agreement across independent teams and datasets would be.

3.3 International evidence used for benchmarking

Chile’s design sits between two poles that the international literature has studied extensively. At one pole, the U.S. Paycheck Protection Program offered loans that were fully guaranteed and largely forgivable. Studies of the PPP document short-run gains in employment, firm survival, and commercial mortgage repayment (Denes et al., 2021; Autor et al., 2022b; Chodorow-Reich et al., 2022; Dalton, 2023; Agarwal et al., 2024), but also find these effects modest relative to fiscal outlay, with a substantial share of support failing to reach intended targets (Autor et al., 2022a; Granja et al., 2022). Because full coverage and forgiveness mute both bank screening and borrower repayment incentives, the PPP is a useful contrast case rather than a comparator. At the other pole are partial-guarantee schemes operating in normal times (Beck et al., 2010; Honohan, 2010; Lelarge et al., 2010; Brown and Earle, 2017; Mullins and Toro, 2018; Bertoni et al., 2023; De Blasio et al., 2018), which study additionality and screening without the confound of a systemic shock.

Chile’s pandemic scheme belongs to a third category: partial, priced, capped guarantees deployed at crisis scale through private intermediaries. The closest published analogues are the euro-area schemes (Altavilla et al., 2025, 2023; Core and De Marco, 2024; Cuciniello, 2024; Nicolas et al., 2025; Mateus and Neugebauer, 2025), the Japanese programs (Hoshi

Table 2: Studies of FOGAPE and FOGAPE-COVID reviewed

Study	Period	Design	Main outcome	Horizon	Status
Cowan et al. (2015)	pre-COVID	Within bank-borrower, insured vs. uninsured loans	Missed payments; default; credit volume	24 months	Peer reviewed (<i>JBF</i>)
Rishmawi and Rojas (2026)	COVID	Approved vs. applied-and-not-approved	Delinquency; internal risk rating	>2 years	Peer reviewed (<i>LAJCB</i>)
Huneeus et al. (2025)	COVID	Descriptive/positive; applications, approvals, tax records; calibrated model	Selection, leverage, aggregate expected loss	2020	Unpublished (R&R, <i>A EJ: Macro</i>)
Chittaro and Sánchez (2026)	COVID	Bank-leniency instrument; dynamic structural model	Cumulative default; welfare	5 years	Unpublished (job market paper)
de Elejalde and Sánchez (2025)	COVID	Discretion in size classification; structural model	Bank reclassification of firms	2020	Unpublished manuscript
Acosta-Henao et al. (2024)	COVID	Sharp RDD at the sales eligibility cut-off	Domestic vs. foreign debt; borrowing costs	Months	NBER WP 33283
Cerda et al. (2023)	COVID	Two randomized loan offers (Chile, Colombia)	Liquidity; earnings	~2 years	IDB working paper
Albagli et al. (2023)	COVID	Descriptive, explicitly non-causal	Sales, employment, credit, leverage	2020–21	Central Bank WP 981
Mullins and Toro (2018)	pre-COVID	Guarantee allocation and bank relationships	New bank relationships	—	Central Bank WP 820

Notes: “Status” records publication status as of August 2026. Several unpublished studies circulate in more than one working paper series; the reference list gives the version consulted.

et al., 2023; Honda et al., 2023), the Swiss facility (Fuhrer et al., 2021), and the UK schemes (Cowling et al., 2022, 2024). Descriptive and institutional accounts of the Spanish and Franco-German schemes (Corredera-Catalán et al., 2021; Massoc, 2022), and cross-country firm-level evidence on pandemic business support more generally (Cirera et al., 2021; Feyen et al., 2021), complete the comparison set. These are drawn on in Section 5 where they speak to a mechanism the Chilean studies leave open.

4 Chile in Comparative Perspective

4.1 How much comparative evidence exists?

A natural way to strengthen design lessons drawn from one country is to corroborate them elsewhere in the region. Latin America deployed public credit guarantee schemes on a large scale during the pandemic, and several of these programs were larger relative to GDP than Chile’s. The systematic search described in Section 3 nonetheless returned a striking result: the peer-reviewed evaluation literature on these programs is close to empty.

For Peru’s Reactiva—the region’s largest scheme—at least eight firm- or loan-level causal evaluations exist (e.g., Burga et al., 2023; Mendoza et al., 2022), but only one has completed peer review (Pardo-Figueroa, 2026); the remainder circulate as central bank, supervisory, or academic working papers. For Brazil, no peer-reviewed firm-level econometric evaluation of PRONAMPE or PEAC-FGI was located; the causal work is a development bank’s evaluation of its own program, a federal audit report, an institutional bulletin, and a master’s dissertation, all of which use variants of matching plus differences-in-differences on the same employment registry and are therefore not independent of one another. Peer-reviewed Brazilian work on pandemic credit exists (Norden et al., 2021), but it studies the policy response broadly rather than the guarantee funds specifically. For Colombia, four credible causal studies of the pandemic support measures exist, and all four remain working papers; the one peer-reviewed evaluation of the Colombian guarantee fund (Arráiz et al., 2014) predates the pandemic. The best regional comparative source, Rudolph et al. (2022), is a descriptive survey of guarantee-scheme administrators in twelve countries; it is a valuable inventory of program parameters, but it has no microdata, no counterfactual, and cannot be read as evidence of program effects. The only published, micro-identified, cross-country evidence on COVID guarantee schemes is from the euro area (Altavilla et al., 2025).

This matters for how the present review should be read. A synthesis of Latin American guarantee programs comparable in rigor to what exists for Chile cannot currently be written,

because the underlying evaluations have not been produced. What can be done, and is done here, is to state Chile’s design precisely, compare it to regional peers on parameters that are documented, and use the international published literature to discipline the mechanisms. The comparative material below therefore serves to bound external validity, not to establish it.

4.2 Design parameters across programs

Table 3 compares Chile’s scheme with Peru’s Reactiva, the two regional programs for which design parameters and ex-post outcomes are documented in primary sources. The contrast is instructive because the two programs made opposite choices on almost every lever while pursuing the same objective.

Two comparisons are worth drawing out. First, Peru combined much higher coverage (up to 98%) with a market mechanism for setting the price, whereas Chile combined lower coverage with an administrative price cap and an explicit first-loss deductible. Chile thus retained more lender risk exposure by design; Peru retained more price discipline. Second, take-up was substantially higher in Chile despite the smaller subsidy per loan, which cautions against reading coverage generosity as the principal determinant of reach. Huneeus et al. (2025) report take-up of roughly 24% of eligible firms in Chile against 14% in Peru, 16% in Colombia, 23% in Mexico, and 16% in Italy.

4.3 Allocation drift across the region

Rudolph et al. (2022) document a pattern that recurs across regional schemes and that bears directly on Section 5: guarantee allocation drifted toward larger firms as the programs scaled. Six of the countries surveyed reduced the share of operations involving micro and small firms in 2020q4 relative to 2019, with the largest shifts in Brazil, Chile, and Mexico (declines of 30, 11, and 7 percentage points, respectively). Measured in guarantee *volume* rather than operation counts, the shifts were far larger: 48 percentage points in Brazil, 65 in Chile, and 14 in Mexico. Chile’s is the largest volume shift the report records.

The same source documents why applications failed in Chile. Between June 5 and December 25, 2020, about 18% of requested operations were rejected. Of micro and small firms’ rejected requests, 36% failed to meet program requirements, while 63% were rejected for non-compliance with the lending institution’s own internal credit risk policies; for medium-sized firms the latter share was 80%. In other words, for the large majority of rejected applications the binding constraint was not the program’s eligibility rules but the banks’ own risk appetite. This is a useful corrective to a framing in which the state sets eligibility and

Table 3: Design of two Latin American pandemic guarantee schemes

	Chile: FOGAPE-COVID	Peru: Reactiva Perú
Guarantee coverage	85% micro/small; 80% medium; 70% large I; 60% large II. Declining in <i>firm size</i> .	98%, 95%, 90%, 80%. Declining in <i>loan size</i> (tranches above S/100,000, S/1m, S/10m, S/20m).
First-loss retention	Deductible on each size-specific portfolio: 5%/3.5%/2.5%, standardized to 2.5% after June 30, 2020.	No portfolio deductible documented in the sources consulted.
Loan pricing	Administrative cap: monetary policy rate + 3% (3.5% in 2020).	Set by competitive reverse auction; institutions bid rates for central bank repo funding.
Quantity cap	One quarter of the firm’s 2019 annual sales.	Tranche ceilings by loan size.
Maturity / grace	24–48 months; grace period of at least 6 months; working capital only.	Medium-term with grace; subsequently extended through two official reprogramming rounds.
Funding	Fund capitalization (USD 3bn); lending on bank balance sheets, alongside a separate central bank facility (FCIC).	Central bank repo operations collateralized by the government guarantee.
Scale (disbursed)	4.6% of GDP.	S/ 58bn placed of S/ 60bn authorized, 7.9% of GDP.
Take-up	~24% of eligible firms by December 2020.	~14% of eligible firms.
Realized guarantee calls	Not yet published in a comparable series.	9.7% of amount lent as of January 2025.

Notes: Chilean parameters from Law 21,229 and [de Elejalde and Sánchez \(2025\)](#); Chilean disbursement and take-up from [Huneus et al. \(2025\)](#). Peruvian parameters, scale, and ex-post guarantee calls from [Banco Central de Reserva del Perú \(2025\)](#); Peruvian take-up as reported in [Huneus et al. \(2025\)](#). Take-up figures are shares of eligible firms and are not strictly comparable across countries because eligibility definitions differ. Blank or qualified cells indicate that no primary source was located, not that the feature was absent.

banks execute: over the relevant range, lenders’ internal policies did most of the allocating.

5 The Evidence, Organized by Design Lever

This section reviews what the studies establish about five design levers. For each, it distinguishes findings on which the literature agrees, findings that are contested, and questions that remain unmeasured. Publication status is stated at the point of use.

5.1 Lever 1: targeting and eligibility

What the evidence agrees on. The exclusion margin does more fiscal work than the inclusion margin. [Huneus et al. \(2025\)](#) estimate that lifting the requirement that borrowers not be in recent default would have expanded guaranteed credit from 3.6% to 7.6% of GDP while raising expected credit losses from 3.7% to 9.9% of baseline credit, a factor of 2.7. By contrast, tightening the loan-size cap from three months to one month of sales would have reduced expected losses to 2.5%. Screens that exclude already-distressed borrowers, and caps that limit exposure per borrower, are thus the two parameters with the largest leverage over the fiscal cost of the program.

There is also agreement that size is a poor proxy for the returns to support. [Chittaro and Sánchez \(2026\)](#) find that young firms generate the highest welfare gains per fiscal peso yet were approximately 41 percentage points less likely to be approved than older firms, because their growth depends on leverage and their measured default risk is correspondingly higher. Age is administratively observable and would be simple to incorporate into a guarantee schedule; the authors estimate that a budget-neutral reallocation of coverage toward younger firms would raise both aggregate welfare and the number of approvals by around 4%. [Andini et al. \(2022\)](#) show for Italy that machine-learning targeting of public guarantees using administrative predictors is feasible and improves on rule-of-thumb allocation, which suggests the Chilean proposal is implementable rather than merely notional.

What is contested. Whether the program in fact reached the firms that needed it most is unresolved, and the disagreement is instructive. On one reading, allocation was efficient: banks used private information to lend to viable firms. On another, allocation drifted sharply toward larger borrowers—[Rudolph et al. \(2022\)](#) record a 65 percentage point fall in the share of guarantee volume going to micro and small firms between 2019 and 2020q4, the largest such shift in their sample—and the drift was driven by lenders’ internal credit policies rather than by program rules.

The randomized evidence complicates both readings. [Cerdea et al. \(2023\)](#) ran two randomized loan offers, in Chile ($n = 10,072$) and Colombia ($n = 3,079$), to small businesses most of which had previously applied for pandemic credit and been rejected on the banks' risk assessment. Take-up was 27% and 29%. The offers raised total liquidity in the formal banking system by 15.7% and commercial liquidity by 26.4%, both significant at the 1% level. But the intention-to-treat effect on firm earnings is indistinguishable from zero throughout the post-period, including after the program ended, even though treatment-on-the-treated estimates are positive and significant. Extending credit at the previously-rejected margin therefore delivered liquidity reliably and measurable earnings gains only ambiguously.

What remains unmeasured. No study evaluates an age-based or model-based targeting rule against the rule actually used; the evidence for retargeting is a model-based counterfactual within a single unpublished paper. Nor does any study identify the counterfactual allocation banks would have chosen absent the size-dependent coverage schedule. The related question of whether broader eligibility improves outcomes has been addressed for Italy ([Lagazio et al., 2021](#)) but not for Chile.

5.2 Lever 2: risk-sharing and pricing

What the evidence agrees on. Partial coverage combined with an explicit first-loss deductible materially reallocates risk to lenders. [Huneus et al. \(2025\)](#) estimate that under the program as designed banks bore about 59% of expected losses and the government 41%; removing the deductible alone would have reversed the split, pushing the government's share to roughly 77%. The deductible is a small-sounding parameter—2.5% of a size-specific portfolio after June 2020—with first-order consequences for who bears risk.

There is also agreement that the interest rate cap did real work. The same study's calibrated model implies that the cap made lending to the riskiest borrowers unprofitable, limiting adverse selection even as guarantees expanded supply; the estimated expected default rate on the guaranteed portfolio, 9.8%, exceeds the 3.5% cap, so absent the cap lenders could have priced for substantially riskier borrowers. Aggregate expected loss on the guaranteed portfolio was estimated at 0.27% of GDP for formal firms, rising to 0.45% when natural persons borrowing as firms are included, or 1.07–1.54% of bank equity.

What is contested. Two assumptions embedded in Chile's schedule are challenged by evidence from other programs.

The first is the level of coverage. [Cerulli and Ventura \(2021\)](#) estimate a dose-response relationship between guarantee coverage and program effectiveness for Italian firms and find

an inverted-U with a maximum around 70% coverage, no significant effects below about 55%, and none above about 80%. FOGAPE-COVID’s coverage for micro and small firms was 85%—above the range in which the Italian evidence detects effects—though the deductible reduced effective coverage.⁵ Yoshino and Taghizadeh-Hesary (2019) reach a similar conclusion for Asian schemes, Uesugi et al. (2010) document Japan’s deliberate move from 100% to 80% coverage, and Arping et al. (2010) show theoretically why coverage generous enough to eliminate lender risk dominates the alternative of co-funding only under restrictive conditions. None of this establishes that 85% was wrong for a systemic shock, where speed may justify generosity; but the review should not describe the Chilean schedule as calibrated to an optimum, because no study has estimated the coverage–effectiveness relationship for Chile.

The second is whether an administrative rate cap binds. Ornelas et al. (2024) show for Brazil that lenders with market power circumvent regulated rate ceilings on subsidized loans through cross-product pricing—raising the price of other products sold to the same client and thereby capturing part of the subsidy. Their setting is pre-pandemic earmarked credit rather than a guarantee scheme, so the result does not transfer directly. It does, however, identify a leakage channel that no Chilean study has tested, and it implies that observed compliance with a rate cap is not sufficient evidence that the subsidy reached borrowers.

What remains unmeasured. How much additional credit a marginal point of coverage buys has not been estimated for Chile. Cowan et al. (2015) provide the closest domestic estimate for the pre-pandemic scheme: one additional dollar of guarantees increased total SME credit by USD 0.65. For Peru, the tranche discontinuities at 98/95/90/80% coverage offer identification of exactly this elasticity and have been exploited in supervisory working papers (Mendoza et al., 2022), but no equivalent exercise exists for Chile.

5.3 Lever 3: bank incentives, monitoring, and risk shifting

This is the lever on which the Chilean literature disagrees most sharply, and the disagreement is substantive rather than semantic.

The disagreement. Cowan et al. (2015), studying the standing scheme in normal times, find that firms selected into the guarantee program were 1.17% more likely to default than similar firms borrowing without guarantees, and interpret this as *adverse* selection. Chittaro and Sánchez (2026) find the opposite sign for the pandemic program: banks used private, relationship-specific information to approve firms that would have defaulted less even without the guarantee. In their decomposition, of the 4.8 percentage point gap in default between

⁵de Elejalde and Sánchez (2025) illustrate the arithmetic: an 85% guarantee with a 2.5% deductible and a 20% portfolio default rate implies an effective coverage rate of 75%.

approved and rejected applicants, 0.8 points are explained by observable characteristics, 1.5 points by banks' private information, and 2.5 points by the causal effect of the loan itself.

These findings are reconcilable, and the reconciliation is a design lesson. The pandemic program layered on the standing scheme two features that the pre-pandemic program lacked: a binding eligibility screen excluding recent defaulters, and a rate cap that removed the possibility of pricing for risk. Both operate directly on the adverse-selection margin. That the sign of selection appears to flip when those parameters are added is among the strongest available arguments that screens and caps are not administrative details but the mechanism by which a guarantee scheme controls who it attracts. It also implies that evidence from normal-times guarantee schemes should not be used to predict selection under crisis schemes with different rules.

Huneus et al. (2025) supply the intermediate step: riskier firms applied disproportionately, and banks rejected them disproportionately, but the demand effect dominated in aggregate quantities. Selection was therefore adverse on the demand side and advantageous on the supply side, with the net sign depending on which margin one measures.

Moral hazard within the borrower relationship. Cowan et al. (2015) also find that firms holding both guaranteed and non-guaranteed loans were 1.67% more likely to miss payments on the *guaranteed* loans, while being no more likely to default on them. Because this comparison is within firm and within bank, it isolates the incentive effect of the guarantee from any selection. Their reading is that guarantees affect firms' incentives to service particular obligations without affecting long-run performance. This is a more precise and more limited claim than a general moral-hazard result, and it is the appropriate published anchor for this lever.

Intermediary responses to the coverage schedule. Because coverage in Chile varied discontinuously with firm size, banks had a potential incentive to classify firms into more generously covered categories. de Elejalde and Sánchez (2025) examine this margin, exploiting the fact that the program administrator supplied three alternative sales-based size criteria that could disagree. Their descriptive evidence is suggestive: where the three criteria agree, banks overwhelmingly comply with the instructed classification; where the criteria conflict and banks have discretion, classification tilts systematically toward the small-firm category, which carries the higher guarantee rate; and compliance weakens as firms' measured sales approach the discretionary region, with a non-trivial fraction of firms instructed to be classified as medium nonetheless classified as small. Reclassification toward the more generously covered category is more likely for firms with a pre-existing lending relationship with the bank.

Evidence on the same mechanism from other settings. The general proposition that

public guarantees alter bank risk-taking does not rest on the Chilean evidence. [Gropp et al. \(2014\)](#) exploit the removal of government guarantees for German savings banks and find that affected banks reduced credit risk by cutting off their riskiest borrowers—the mirror image of the mechanism at issue. [Carletti et al. \(2023\)](#) show theoretically that guarantees can either tighten or loosen bank underwriting standards depending on coverage and on bank capitalization. [Nicolas et al. \(2025\)](#) document that guaranteed loans generated a regulatory capital windfall for French banks, identifying a motive for guaranteed lending that has nothing to do with borrower quality. [Core and De Marco \(2024\)](#) show that lender technology, not just lender incentives, shapes who obtains guaranteed credit.

What remains unmeasured. No Chilean study tests whether guaranteed lending displaced non-guaranteed credit to the same firm. This is not a minor omission: [Altavilla et al. \(2025\)](#) find using euro-area credit register data that one euro of guaranteed lending was associated with a reduction of 28 cents in non-guaranteed credit to the same firm relative to other banks lending to that firm, with substitution strongest for riskier and smaller borrowers. Regional evidence points the same way: in Peru, banks more exposed to Reactiva expanded guaranteed credit while cutting non-guaranteed lending sharply, and directed guaranteed credit toward their larger incumbent clients even though the delinquency response to credit was largest among small firms in exposed industries ([Burga et al., 2023](#)). If displacement of that magnitude occurred in Chile, estimates of the program’s net effect on credit would be overstated.

5.4 Lever 4: coordination with complementary policies

What the evidence agrees on. The guarantee scheme did not operate alone, and its effects are not separable from the instruments deployed alongside it. [Huneus et al. \(2025\)](#) show that Chile’s employment protection program, though roughly seven times smaller than the credit program in fiscal terms (0.62% versus 4.6% of GDP), materially changed credit outcomes. Take-up of employment protection was unrelated to ex-ante credit risk but strongly related to real-side distress, so participation served as an informative signal that improved bank screening. Firms using both instruments borrowed less. In the study’s counterfactual, removing the employment program would have raised average default probabilities from 7.4% to 8.5%, expanded guaranteed credit from 3.6% to 4.2% of GDP, and increased expected losses from 3.7% to 5.0% of baseline credit.

[Acosta-Henao et al. \(2024\)](#) document a second complementarity, between the guarantee scheme and central bank liquidity provision. Exploiting the sales threshold for eligibility in a regression discontinuity design, they estimate that eligible firms increased their domestic debt

share by 9.4 percentage points relative to ineligible firms at the cutoff. Before the pandemic, Chilean firms faced an uncovered interest parity premium of about 3.95 percentage points on domestic-currency borrowing; during the crisis this premium disappeared for policy-eligible firms while persisting for ineligible ones, driven by falling domestic rates rather than by changes in foreign funding costs. Because eligible and ineligible firms often borrowed from the same lenders, the mechanism operates through how banks priced risk rather than through reallocation of firms across banks.

What is contested. Whether wage subsidies improve credit outcomes is not settled outside Chile. [Gómez and Rodríguez-Novoa \(2024\)](#) study Colombia’s payroll subsidy, credit guarantees, and a debt moratorium jointly on the same firms using the credit registry, and find that the payroll subsidy had no detectable effect on credit payment behavior, in contrast to the Chilean result that employment support reduced credit risk. They also document a mechanism absent from every Chilean study: banks systematically re-rated program participants as riskier once support ended, which implies that participation in relief programs can carry a stigma cost that materializes at exit. [Bonilla-Mejía et al. \(2025\)](#) find that the same Colombian program did raise employment, so the divergence concerns the credit margin specifically rather than the program’s effectiveness overall.

What remains unmeasured. No study jointly estimates the effects of the guarantee scheme, the central bank facility, and the employment program in Chile within a single design. Each existing study holds the others fixed or absorbs them into controls, so the interaction terms that would tell a policymaker how to size one instrument given another are not identified.

5.5 Lever 5: repayment and exit

The central disagreement. Whether guaranteed crisis credit prevents defaults or defers them is the question on which the Chilean evidence divides, and it is the question with the largest implications for how such a program should be wound down.

[Rishmawi and Rojas \(2026\)](#) compare firms that accessed FOGAPE-COVID with firms that sought the credit but did not obtain it. They find that access reduced the probability of delinquency by about 1 percentage point in the short run and 4 percentage points in the medium run, that the effect persists for more than two years after credit allocation, that it is also negative relative to eligible firms that did not apply, that it is concentrated among micro and small firms, and that access reduced the probability of a deterioration in the internal risk classification banks assign.

Chittaro and Sánchez (2026) instrument approval with bank-level approval leniency interacted with pre-crisis firm–bank relationships, and track outcomes for five years. They estimate reductions in cumulative default of 2.5 percentage points in 2020 and 5.2 in 2021, after which the estimates lose significance and the sign eventually turns. Their conclusion is that the program postponed defaults for about two years without reducing them in the long run.

Cowan et al. (2015) supply the normal-times benchmark rather than a third estimate of the crisis program: for the standing pre-pandemic scheme, they find that guarantees worsened payment behavior on guaranteed loans while leaving long-run default unaffected. The result does not speak to FOGAPE-COVID directly, but it establishes the direction in which the guarantee itself pushes repayment incentives when no crisis-specific screen or rate cap is in force.

The two crisis-program estimates are not mutually inconsistent once the estimands are separated. They use different comparison groups (applicants who were not approved, versus the marginal borrower at a lenient rather than a strict bank), different outcomes (delinquency and risk-rating deterioration, versus cumulative default), and different horizons. A demand-conditioned contrast between approved and rejected applicants asks whether receiving the loan helped those who wanted it; a lender-leniency instrument recovers an effect for borrowers near the approval margin, who are precisely the firms most likely to be fragile when repayment begins. It would be entirely coherent for the program to reduce delinquency for the average recipient over two years while failing to reduce cumulative default over five years for marginal recipients. Distinguishing these is not a technicality: a policymaker deciding whether to extend maturities needs the marginal-borrower estimate, while one assessing whether the program was worth its cost needs the average. The normal-times benchmark then supplies a third piece of information: the same guarantee fund, operating without the crisis program’s screen and cap, moved payment behavior in the opposite direction—which is what one would expect if those design features, rather than the guarantee itself, govern the sign.

What the disagreement does establish is that confident statements in either direction are not currently warranted, and that the horizon over which the Chilean program is judged is doing much of the work in existing assessments.

What the regional record adds. Peru offers the only complete ex-post accounting in Latin America. As of January 2025, guarantee calls under Reactiva totaled S/ 5,635 million, or 9.7% of the amount lent, with 87% of program credit repaid (Banco Central de Reserva del Perú, 2025). Adding overdue loans brings the figure to 10.8% of program credit, against a delinquency ratio of 21.8% for comparable borrowers who did not receive Reactiva credit. On

these aggregates the program’s realized fiscal cost was well below both its ex-ante provisioning and the loss rate on comparable unguaranteed lending.

The same source, however, documents a pattern directly relevant to exit design. Measured by number of firms, borrowers who took the first official reprogramming—that is, those granted additional grace—defaulted at 28.7%, against 18.9% for firms that never reprogrammed and 2.6% for those reprogramming in the second round. This is a selection contrast, not a causal estimate: firms needing extended grace were already the weakest. But it is precisely the pattern one would expect if extending maturities sorts distress forward rather than resolving it, and it is the clearest regional evidence that the design of the repayment phase is consequential.

What the international record adds. The concern that guaranteed crisis credit sustains non-viable firms has been examined directly elsewhere. [Hoshi et al. \(2023\)](#) find that Japan’s COVID business support programs increased the share of firms meeting zombie criteria. [Zoller-Rydzek and Keller \(2020\)](#) argue that Swiss guaranteed loans risked keeping unproductive firms alive. [Faria-e Castro et al. \(2024\)](#) document evergreening incentives when lenders can roll over troubled credit. [Demmou and Franco \(2021\)](#) analyze the productivity consequences of the transition from hibernation to reallocation. [Cowling et al. \(2022, 2024\)](#) model default on the UK Bounce Back Loan Scheme, whose full coverage removed lender screening entirely and which experienced markedly worse outcomes. [Bajo et al. \(2026\)](#) examine financial recovery among Italian recipients of state-guaranteed loans. This literature does not adjudicate the Chilean disagreement, but it establishes that the repayment phase is where guarantee schemes have historically generated their costs, and that programs preserving lender risk retention have generally fared better than those that did not.

What remains unmeasured. Chile has no published analysis of realized guarantee calls. The fund’s contingent liabilities are reported annually by the Budget Office ([Dirección de Presupuestos \(DIPRES\), 2025](#)), and the CMF holds the loan-level performance data, but no study has linked the realized payout series to the ex-ante loss estimates in [Huneus et al. \(2025\)](#). That comparison is the single most informative exercise still available on this program: it would adjudicate between the competing default findings using outturns rather than projections.

6 Limitations and Scope Conditions

The conclusions drawn here are bounded in several respects, and the bounds are wide enough to state explicitly.

One country, one program, one shock. The evidence concerns a single guarantee scheme in a single country during a single crisis episode. The COVID-19 shock had features—a common, exogenous, and widely anticipated-to-be-temporary cause; simultaneous deployment of wage subsidies and central bank liquidity; and unusually strong fiscal capacity in Chile relative to regional peers—that will not characterize the next crisis. Design lessons that depend on the shock being temporary and correlated across firms, in particular the case for broad eligibility and back-loaded repayment, are the least transferable.

An evidence base that is small and largely unreviewed. Roughly a dozen studies evaluate the program, several sharing data and authors. Only two have completed peer review, and a third is under revision at a journal. Publication status is recorded study-by-study in Table 2; readers should weight findings accordingly.

Formal firms only. The administrative data underlying every Chilean study cover formal firms with bank relationships. Chittaro and Sánchez (2026) note that their sample does not capture informal firms or very early-stage startups, and Huneus et al. (2025) make the same point. In economies where informality is more prevalent than in Chile, a bank-intermediated guarantee scheme mechanically cannot reach a large share of firms, and the Chilean results say nothing about that margin.

Model dependence of the welfare and fiscal numbers. The welfare estimates come from a calibrated dynamic model whose counterfactuals are not directly observed (Chittaro and Sánchez, 2026). The aggregate expected-loss figures assume zero recovery and abstract from endogenous responses by firms and banks (Huneus et al., 2025), which their authors describe as making the estimates a lower bound in some dimensions and an overstatement of realized fiscal burden in others. Neither set of numbers should be read as a measured outturn.

Identification caveats in the individual designs. The sudden-stop results are identified from a sharp discontinuity in *eligibility*, not in treatment (Acosta-Henao et al., 2024); the authors note that only about fifteen firms near the cutoff actually took guaranteed loans, which is why a fuzzy design was not estimated. Estimates of the program’s effect on default are sensitive to the inclusion of pre-crisis bank-assessed risk controls (Chittaro and Sánchez, 2026). And their structural model abstracts from the size-specific guarantee schedule, assuming a uniform coverage share.

External validity of the comparative material. As Section 4 documents, the regional comparison rests on program parameters and one national ex-post accounting, not on comparable causal evaluations, because those evaluations have not been produced. The cross-country contrasts in this paper should be read as generating hypotheses about design, not as testing them.

7 Conclusions

Chile’s pandemic guarantee scheme is among the most thoroughly studied crisis credit interventions in any emerging economy, and reading its evaluations together yields conclusions that none of them states individually.

Three findings are robust across the corpus. First, the parameters that govern fiscal exposure are the exclusion screen and the quantity cap, not the coverage rate. Relaxing the requirement that borrowers not be in recent default would have nearly tripled expected losses, while halving the loan-size cap would have cut them by a third. Second, first-loss retention is what determines who bears risk: a deductible of 2.5% was the difference between the government absorbing 41% and 77% of expected losses. Third, guarantee schemes are not separable from the instruments deployed beside them—in Chile, employment protection both reduced firms’ borrowing needs and gave banks an informative signal about real-side distress, and its removal would have raised expected credit losses by roughly a third.

Two conclusions that the previous literature treated as settled should instead be treated as open. The first concerns defaults. The claim that FOGAPE-COVID postponed rather than prevented them rests on one unpublished study using a marginal-borrower design; a peer-reviewed study using a demand-conditioned comparison finds reductions in delinquency persisting beyond two years; and the peer-reviewed evidence on the pre-pandemic scheme, operating in normal times, finds guarantees worsening payment behavior. These are different estimands under different design regimes, and Section 5.5 argues they can all be correct. But the policy question—whether the program bought time or bought solvency—is not currently answered. The second concerns bank behavior. The evidence that banks reclassified firms to obtain more generous coverage comes from a single unpublished manuscript whose main-text results are descriptive.

The regional picture is sobering. Latin America deployed guarantee schemes worth several points of GDP, and the peer-reviewed evaluation literature on them consists of a single article. Where evaluations exist they are overwhelmingly grey, frequently share data and authors, and rely on weaker identification than the Chilean work. Any claim that a design lesson from Chile transfers to Peru, Brazil, or Colombia is at present a conjecture.

Four pieces of work would resolve most of what this review cannot. The first is to link realized guarantee payouts to ex-ante loss projections. Chile publishes contingent liabilities annually and the CMF holds loan-level performance data; comparing outturns to projections would adjudicate the default disagreement using outcomes rather than models, and requires no new data collection. The second is to test whether guaranteed lending displaced non-

guaranteed credit to the same firm. Euro-area evidence finds that a euro of guaranteed lending displaced 28 cents of unguaranteed credit; the Chilean credit registry can run this test. The third is to evaluate targeting rules against the rule actually used—the case for age-based coverage currently rests on a model-based counterfactual, and Italian evidence suggests administrative-data targeting is implementable. The fourth is replication outside Chile, particularly of the reclassification result, which is the finding with the sharpest supervisory implications.

The broader lesson is about how crisis credit should be assessed. The acute-phase metrics—speed, take-up, volumes—were satisfied comfortably, and the studies agree on them. The metrics that matter for whether the intervention was worth its cost concern the repayment phase, and there the Chilean evidence is thinnest and the international record most cautionary. Programs designed to be deployed in weeks should be designed at the same time to be unwound over years; on current evidence, that second design problem received far less attention than the first.

Declaration Statement

Funding: This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used GitHub Copilot, ChatGPT (OpenAI), and Claude (Anthropic) in order to search for and verify the bibliographic references cited in this text, to draft and revise portions of the manuscript, to construct the summary tables, and to check the quantitative claims reported here against their original sources, as well as to check grammar and spelling. These tools were not used to generate or analyze data. All substantive judgments—the selection of studies reviewed, the interpretation of their findings, and the conclusions drawn from them—are the author’s own. After using these tools/services, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

- ACOSTA-HENAO, M., A. FERNÁNDEZ, P. GOMEZ-GONZALEZ, AND S. KALEMLI-OZCAN (2024): “Firm Financing During Sudden Stops: Can Governments Substitute Markets?” NBER Working Paper 33283, National Bureau of Economic Research.
- AGARWAL, S., B. W. AMBROSE, L. A. LOPEZ, AND X. XIAO (2024): “Did the Paycheck Protection Program Help Small Businesses? Evidence from Commercial Mortgage-Backed Securities,” *American Economic Journal: Economic Policy*, 16, 95–132.
- ALBAGLI, E., A. FERNÁNDEZ, J. GUERRA-SALAS, F. HUNEEUS, AND P. MUÑOZ (2023): “Anatomy of Firms’ Margins of Adjustment: Evidence from the COVID Pandemic,” Documentos de Trabajo del Banco Central de Chile 981, Banco Central de Chile, descriptive, explicitly non-causal.
- ALTAVILLA, C., F. BARBIERO, M. BOUCINHA, AND L. BURLON (2023): “The Great Lockdown: Pandemic response policies and bank lending conditions,” *European Economic Review*, 156, 104478.
- ALTAVILLA, C., A. ELLUL, M. PAGANO, A. POLO, AND T. VLASSOPOULOS (2025): “Loan guarantees, bank lending and credit risk reallocation,” *Journal of Financial Economics*, 172, 104137.
- ANDINI, M., M. BOLDRINI, E. CIANI, G. DE BLASIO, A. D’IGNAZIO, AND A. PALADINI (2022): “Machine learning in the service of policy targeting: The case of public credit guarantees,” *Journal of Economic Behavior & Organization*, 198, 434–475.
- ARPING, S., G. LÓRÁNT, AND A. D. MORRISON (2010): “Public initiatives to support entrepreneurs: Credit guarantees versus co-funding,” *Journal of Financial Stability*, 6, 26–35.
- ARRÁIZ, I., M. MELÉNDEZ, AND R. STUCCHI (2014): “Partial credit guarantees and firm performance: evidence from Colombia,” *Small Business Economics*, 43, 711–724.
- AUTOR, D., D. CHO, L. D. CRANE, M. GOLDAR, B. LUTZ, J. MONTES, W. B. PETERMAN, D. RATNER, D. VILLAR, AND A. YILDIRMAZ (2022a): “The \$800 Billion Paycheck Protection Program: Where Did the Money Go and Why Did It Go There?” *Journal of Economic Perspectives*, 36, 55–80.
- (2022b): “An evaluation of the Paycheck Protection Program using administrative payroll microdata,” *Journal of Public Economics*, 211, 104664.

- BAJO, E., M. BARBI, AND M. BIGELLI (2026): “Back From the Brink: State-Guaranteed Loans and Financial Recovery,” *European Financial Management*, 32, 140–155.
- BANCO CENTRAL DE RESERVA DEL PERÚ (2025): *Reactiva: Diseño y resultados*, Lima: Banco Central de Reserva del Perú, edited by Carlos Montoro, José Luis Bustamante and Marco Vega. Institutional report.
- BECK, T., L. F. KLAPPER, AND J. C. MENDOZA (2010): “The typology of partial credit guarantee funds around the world,” *Journal of Financial Stability*, 6, 10–25, publisher: Elsevier.
- BERTONI, F., M. G. COLOMBO, AND A. QUAS (2023): “The long-term effects of loan guarantees on SME performance,” *Journal of Corporate Finance*, 80, 102408.
- BONILLA-MEJÍA, L., L. A. FLOREZ, D. HERMIDA, F. LASSO-VALDERRAMA, L. F. MORALES, AND J. PULIDO (2025): “Payroll Subsidies for SMEs in Informal Labor Markets,” Borradores de Economía 1338, Banco de la República.
- BROWN, J. D. AND J. S. EARLE (2017): “Finance and growth at the firm level: Evidence from SBA loans,” *The Journal of Finance*, 72, 1039–1080, publisher: Wiley Online Library.
- BURGA, C., W. CUBA, E. DÍAZ, AND E. SÁNCHEZ (2023): “Loan Guarantees and Bank Incentives: Evidence from Covid-19 Relief Funds in Peru,” Working Paper 2023-001, Banco Central de Reserva del Perú.
- CARLETTI, E., A. LEONELLO, AND R. MARQUEZ (2023): “Loan guarantees, bank underwriting policies and financial stability,” *Journal of Financial Economics*, 149, 260–295.
- CERDA, M., P. GERTLER, S. HIGGINS, A. M. MONTOYA, E. PARRADO, AND R. UNDURRAGA (2023): “The Causal Impact of Covid-19 Government-backed Loans on MSMEs Liquidity and Earnings,” Idb working paper series, Inter-American Development Bank, randomized experiments in Chile and Colombia.
- CERULLI, G. AND M. VENTURA (2021): “A dose–response approach to evaluate the effects of different levels of partial credit guarantees,” *Applied Economics*, 53, 1418–1434.
- CHITTARO, L. AND C. SÁNCHEZ (2026): “Selection in Crisis Lending: Evidence from Chile’s Government-Guaranteed Loans,” Job market paper, Stanford University. This version: January 12, 2026.

- CHODOROW-REICH, G., B. IVERSON, AND A. SUNDERAM (2022): “Lessons learned from support to business during COVID-19,” *The Hamilton Project, Brookings Institution, Washington, DC*.
- CIRERA, X., M. CRUZ, E. DAVIES, A. GROVER, L. IACOVONE, J. E. LÓPEZ CÓRDOVA, D. MEDVEDEV, F. O. MADUKO, G. NAYYAR, S. REYES ORTEGA, AND J. TORRES (2021): “Policies to Support Businesses through the COVID-19 Shock: A Firm-Level Perspective,” *The World Bank Research Observer*, 36, 41–66.
- CORE, F. AND F. DE MARCO (2024): “Information Technology and Credit: Evidence from Public Guarantees,” *Management Science*, 70, 6202–6219.
- CORREDERA-CATALÁN, F., F. DI PIETRO, AND A. TRUJILLO-PONCE (2021): “Post-COVID-19 SME financing constraints and the credit guarantee scheme solution in Spain,” *Journal of Banking Regulation*, 22, 250–260.
- COWAN, K., A. DREXLER, AND A. YAÑEZ (2015): “The effect of credit guarantees on credit availability and delinquency rates,” *Journal of Banking & Finance*, 59, 98–110.
- COWLING, M., N. WILSON, P. NIGHTINGALE, AND M. KACER (2022): “Predicting future default on the Covid-19 bounce back loan scheme: The £46.5 billion question,” *International Small Business Journal*, 40, 650–666.
- (2024): “The hazards of delivering a public loan guarantee scheme: An analysis of borrower and lender characteristics,” *International Small Business Journal*, 42, 212–245.
- CUCINIELLO, V. (2024): “Credit allocation to businesses in Italy amid the Covid-19 crisis,” *Economics Letters*, 238, 111724.
- DALTON, M. (2023): “Putting the Paycheck Protection Program into Perspective: An Analysis Using Administrative and Survey Data,” *National Tax Journal*, 76, 393–437, publisher: The University of Chicago Press.
- DE BLASIO, G., S. DE MITRI, A. D’IGNAZIO, P. F. RUSSO, AND L. STOPPANI (2018): “Public guarantees to SME borrowing. A RDD evaluation,” *Journal of Banking & Finance*, 96, 73–86, publisher: Elsevier.
- DE ELEJALDE, R. AND C. SÁNCHEZ (2025): “Risk-shifting Incentives Under Government Credit Guarantees,” Unpublished manuscript, Central Bank of Chile. This version: February 25, 2025.

- DEMMOU, L. AND G. FRANCO (2021): “From hibernation to reallocation: loan guarantees and their implications for post-COVID-19 productivity,” OECD Economics Department Working Papers 1687, OECD Publishing.
- DENES, M., S. LAGARAS, AND M. TSOUTSOURA (2021): “First Come, First Served: The Timing of Government Support and Its Impact on Firms,” SSRN Working Paper 3845046, Social Science Research Network, Rochester, NY.
- DIRECCIÓN DE PRESUPUESTOS (DIPRES) (2025): “Informe de Pasivos Contingentes 2025,” Tech. rep., Ministerio de Hacienda, Gobierno de Chile, official fiscal report.
- FARIA-E CASTRO, M., P. PAUL, AND J. M. SÁNCHEZ (2024): “Evergreening,” *Journal of Financial Economics*, 153, 103778.
- FEYEN, E., T. ALONSO GISPERT, T. KLIATSKOVA, AND D. S. MARE (2021): “Financial Sector Policy Response to COVID-19 in Emerging Markets and Developing Economies,” *Journal of Banking & Finance*, 133, 106184.
- FUHRER, L. M., M.-A. RAMELET, AND J. TENHOFEN (2021): “Firms’ participation in the Swiss COVID-19 loan programme,” *Swiss Journal of Economics and Statistics*, 157, 2.
- GÓMEZ, C. AND D. RODRÍGUEZ-NOVOA (2024): “Firm Support Measures, Credit Payment Behavior, and Credit Risk,” Borradores de Economía 1277, Banco de la República.
- GRANJA, J., C. MAKRIDIS, C. YANNELIS, AND E. ZWICK (2022): “Did the paycheck protection program hit the target?” *Journal of Financial Economics*, 145, 725–761.
- GROPP, R., C. GRUENDL, AND A. GUETTLER (2014): “The Impact of Public Guarantees on Bank Risk-Taking: Evidence from a Natural Experiment*,” *Review of Finance*, 18, 457–488.
- HONDA, T., K. HOSONO, D. MIYAKAWA, A. ONO, AND I. UESUGI (2023): “Determinants and effects of the use of COVID-19 business support programs in Japan,” *Journal of the Japanese and International Economies*, 67, 101239.
- HONOHAN, P. (2010): “Partial credit guarantees: Principles and practice,” *Journal of Financial Stability*, 6, 1–9.
- HOSHI, T., D. KAWAGUCHI, AND K. UEDA (2023): “Zombies, again? The COVID-19 business support programs in Japan,” *Journal of Banking & Finance*, 147, 106421, publisher: Elsevier.

- HUNEEUS, F., J. P. KABOSKI, M. LARRAIN, S. L. SCHMUKLER, AND M. VERA (2025): “Crisis Credit, Employment Protection, Indebtedness, and Risk,” Unpublished manuscript, October 2025; under revision at the American Economic Journal: Macroeconomics. An earlier version circulated as World Bank Policy Research Working Paper 10958.
- LAGAZIO, C., L. PERSICO, AND F. QUERCI (2021): “Public guarantees to SME lending: Do broader eligibility criteria pay off?” *Journal of Banking & Finance*, 133, 106287.
- LELARGE, C., D. SRAER, AND D. THESMAR (2010): “Entrepreneurship and credit constraints: Evidence from a French loan guarantee program,” in *International differences in entrepreneurship*, University of Chicago Press, 243–273.
- MASSOC, E. C. (2022): “Having banks ‘play along’: state-bank coordination and state-guaranteed credit programs during the COVID-19 crisis in France and Germany,” *Journal of European Public Policy*, 29, 1135–1152.
- MATEUS, M. AND K. NEUGEBAUER (2025): “Stayin’ alive? Government support measures in Portugal during the Covid-19 pandemic,” *Portuguese Economic Journal*, 24, 335–372.
- MENDOZA, C., B. GUTIÉRREZ, AND D. ROMERO (2022): “Garantías gubernamentales y la oferta de crédito: El caso Reactiva Perú,” SBS Documentos de Trabajo DT/04/2022, Superintendencia de Banca, Seguros y AFP (SBS).
- MULLINS, W. AND P. TORO (2018): “Credit guarantees and new bank relationships,” Documentos de Trabajo del Banco Central de Chile 820, Banco Central de Chile.
- NICOLAS, T., S. UNGARO, AND E. VANSTEENBERGHE (2025): “Public-Guaranteed Loans, Bank Risk-Taking, and the Regulatory Capital Windfall,” *Journal of Financial Services Research*.
- NORDEN, L., D. MESQUITA, AND W. WANG (2021): “COVID-19, policy interventions and credit: The Brazilian experience,” *Journal of Financial Intermediation*, 48, 100933.
- ORNELAS, J. R. H., A. PEDRAZA, C. RUIZ-ORTEGA, AND T. C. SILVA (2024): “Market Power and the Transmission of Loan Subsidies,” *The Review of Corporate Finance Studies*, 13, 931–965.
- PARDO-FIGUEROA, R. (2026): “The unintended de-dollarization effect of a public credit guarantee program,” *Latin American Journal of Central Banking*, 100205.

- RISHMAWI, J. AND C. ROJAS (2026): “Impact of the access to Fogape-Covid program on firm delinquency: Evidence from commercial loans within the banking system,” *Latin American Journal of Central Banking*, 7, 100189.
- RUDOLPH, H. P., F. A. DIAZ KALAN, AND F. MIGUEL LIRIANO (2022): “Challenges of Public Credit Guarantee Schemes in Latin America during the Pandemic,” Policy Research Working Paper 9895, The World Bank, survey of guarantee-scheme administrators, no microdata.
- UESUGI, I., K. SAKAI, AND G. M. YAMASHIRO (2010): “The Effectiveness of Public Credit Guarantees in the Japanese Loan Market,” *Journal of the Japanese and International Economies*, 24, 457–480.
- YOSHINO, N. AND F. TAGHIZADEH-HESARY (2019): “Optimal credit guarantee ratio for small and medium-sized enterprises’ financing: Evidence from Asia,” *Economic Analysis and Policy*, 62, 342–356.
- ZOLLER-RYDZEK, B. AND F. KELLER (2020): “COVID-19: Guaranteed Loans and Zombie Firms,” *CESifo Economic Studies*, 66, 322–364.